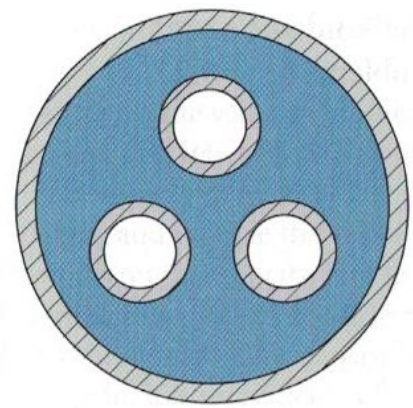
CHEMICAL ENGINEERING 2O04 - Assignment #4

Dr. D.R. Latulippe Due: At 7:00pm on Tue Mar 17th

Notes:

- As per the course outline, full credit is given for completion; thus, your submission will NOT be graded for correctness.
- Your single PDF submission to the Avenue course page should include all your solutions to the questions below.

QUESTION 1



The accompanying figure is the cross-section of a heat-exchanger – the grey-hatched sections indicate the parts made of copper (chosen for its high thermal conductivity) and the solid blue-shaded 'shell' section is where the hotter fluid is flowing. The heat exchanger has the following dimensions:

- the outer shell has an internal diameter (D_i) of 80 mm
- the three tubes on the inside have an external diameter (D_o) of 25 mm
- the thickness of the shell and tubes are 1.5 mm ←EXTRA INFO THAT IS NOT NEEDED!

a) Calculate the Reynolds number when ethylene glycol at 75 °C (*HINT: use PIPE-FLO to find relevant properties*) is the hotter fluid flowing at a volumetric rate of 950 L/min.

From Slide 5 of PDF_08: $Re = \frac{\rho V_{avg} D_h}{\mu}$

From Slide 3 of PDF_08: $D_h = \frac{4A}{P}$

For the blue-shaded section through which the ethylene glycol is flowing:

$$A = \frac{\pi}{4}D_i^2 - 3\frac{\pi}{4}D_o^2$$
$$P = \pi D_i + 3D_o$$

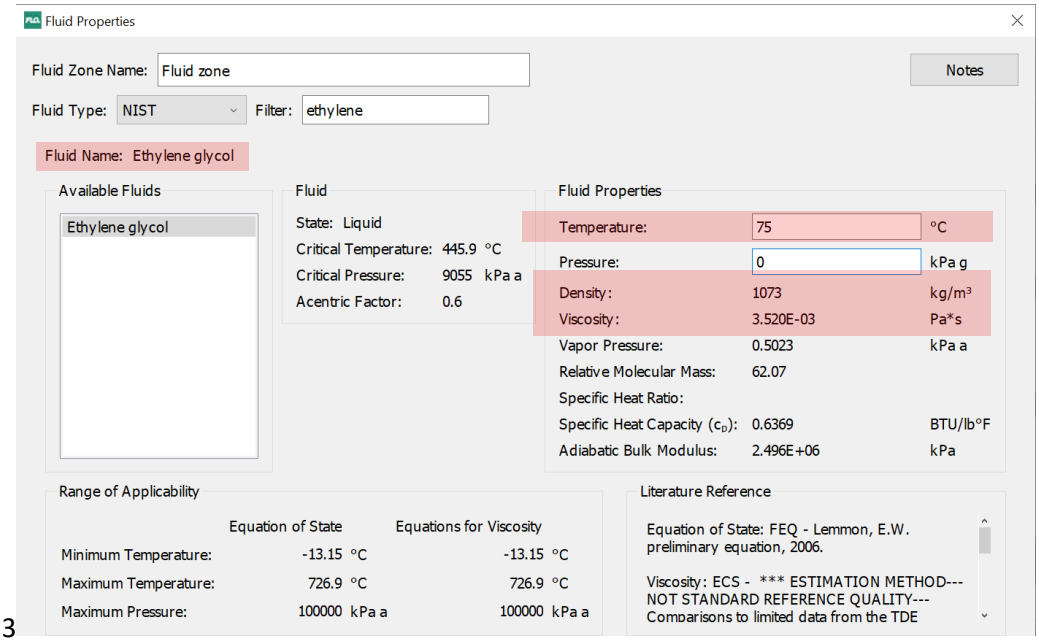
such that $D_h = \frac{4\left(\frac{\pi}{4}D_i^2 - 3\frac{\pi}{4}D_o^2\right)}{\pi D_i + 3D_o}$

substituting in values given above gives $D_h = 0.029m$

Calculate average velocity of ethylene glycol in blue-shaded section:

$$V_{avg} = \frac{Q}{A} = \frac{Q}{\frac{\pi}{4}D_i^2 - 3\frac{\pi}{4}D_o^2} = \frac{950\frac{L}{min} * \frac{1m^3}{1000L} * \frac{1min}{60s}}{\frac{\pi}{4}(0.08m)^2 - 3\frac{\pi}{4}(0.025m)^2} = 4.45\,{}^{m}/_{s}$$

Screenshot from PIPE-FLO:

3 

Back to the equation for Reynolds number: $Re = \frac{\rho V_{avg} D_h}{\mu}$

$$Re = \frac{\left(1073\,{}^{kg}/_{m^3}\right)(4.45\,{}^{m}/_{s})(0.029m)}{3.52e^{-3}Pa \cdot s} = 39{,}647$$

b) Calculate the pressure drop per unit length (kPa/m) in the shell section of a horizontal heat exchanger (i.e. no change in elevation).

$$\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + gz_1 = \frac{P_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + gz_2 + h_l + h_{l_m}$$

Since horizontal $z_1 = z_2$

Since the same cross-sectional geometry $V_1 = V_2$

and

Since no known fittings, valves, etc $h_{l_m} = 0$

Thus, the simplified form of the Bernoulli Equation is:

$$\frac{P_1}{\rho} = \frac{P_2}{\rho} + f\frac{L}{D_h}\frac{V_{avg}^2}{2}$$

Which is rearranged to:

$$\frac{P_1 - P_2}{L} = \rho f \frac{1}{D_h}\frac{V_{avg}^2}{2}$$

Calculate relative roughness (ε):

From Slide 46 of PDF_07, copper roughness = $3.3e^{-6}$ ft = $1e^{-6}$ m

$$\varepsilon = \frac{e}{D_h} = \frac{1\mathrm{e}^{-6}\ \mathrm{m}}{0.029m} = 0.00003$$

Calculate friction factor (f):

Use Haaland Equation with $\varepsilon = 0.00003$ and $Re = 39{,}647$

$$f = \left[\frac{1}{-1.8\log_{10}\left[\left(\frac{0.00003}{3.7}\right)^{1.11} + \frac{6.9}{161{,}514}\right]}\right]^2 = 0.0219$$

Substituting in values given above:

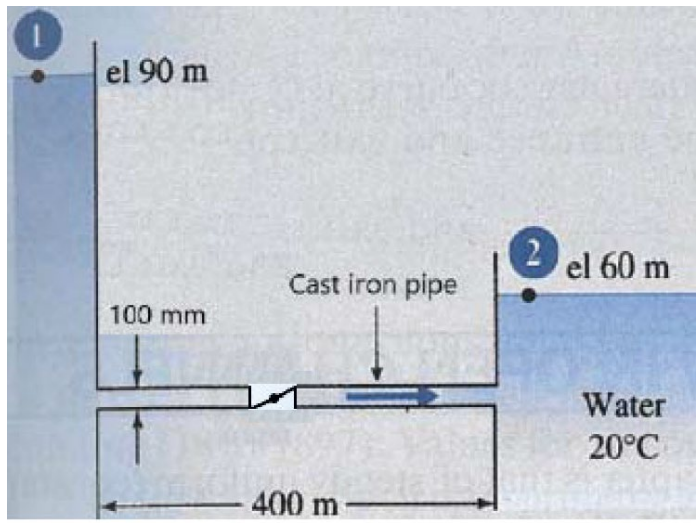
$$\frac{P_1 - P_2}{L} = \left(1073\,{}^{kg}/_{m^3}\right)(0.0219)\frac{1}{0.029m}\frac{(4.45\,{}^{m}/_{s})^2}{2}$$

$$\frac{P_1 - P_2}{L} = 7993\,{}^{Pa}/_{m} * {}^{1kPa}/_{1000Pa}$$

$$\boldsymbol{\frac{P_1 - P_2}{L} = 7.993\,{}^{kPa}/_{m}}$$

QUESTION 2

In the accompanying figure, water is transported between two large reservoirs through a long cast iron pipe. The system is almost exactly the same as that shown on slide 28 of the PDF_08 notes – the one difference is that a butterfly valve was installed in the pipe.



a) Calculate the volumetric flow rate of water (m^3/min) in the pipe when the butterfly valve disc is positioned to θ = 60° (i.e. mostly closed).

Type III problem (so follow the strategy outlined on slide 26 of PDF_08)

1. Tabulate data for minor loss coefficients:

$$K_{square-edged_inlet} = 0.5$$
$$K_{butterfly-valve} = 118$$
$$K_{exit} = 1$$

2. Calculate relative roughness (ε):

From Table 8.1 (on Slide 45 of PDF_07), cast iron roughness = 0.26 mm

$$\varepsilon = \frac{e}{D} = \frac{0.26mm}{100mm} = 0.0026$$

3. Guess value for friction factor (f):

Use Haaland Equation with $\varepsilon = 0.0026$ and very big Re (say 100,000,000)

$$f = \left[\frac{1}{-1.8\log_{10}\left[\left(\frac{0.0026}{3.7}\right)^{1.11} + \frac{6.9}{100{,}000{,}000}\right]}\right]^2 = 0.0252$$

4. Calculate average velocity using Bernoulli Equation:

$$\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + gz_1 = \frac{P_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + gz_2 + h_l + h_{l_m}$$

Based on figure shown above:

$$P_1 = P_2 = P_{atm}$$
$$V_1 \approx 0$$
$$V_2 \approx 0$$

Thus, the simplified form of the Bernoulli Equation is:

$$gz_1 = gz_2 + h_l + h_{l_m}$$

or

$$gz_1 = gz_2 + f\frac{L}{D_h}\frac{V_{avg}^2}{2} + \sum\left[K\frac{V_c^2}{2}\right]$$

$$gz_1 = gz_2 + f\frac{L}{D_h}\frac{V_{avg}^2}{2} + K_{square-edged_inlet}\frac{V_{avg}^2}{2} + K_{butterfly-valve}\frac{V_{avg}^2}{2} + K_{exit}\frac{V_{avg}^2}{2}$$

Rearrange and solve for V_{avg}:

$$V_{avg} = \sqrt{\frac{2g(z_1 - z_2)}{f\frac{L}{D_h} + K_{square-edged_inlet} + K_{butterfly-valve} + K_{exit}}}$$

Substituting in values given above

$$V_{avg} = \sqrt{\frac{2 * 9.81\,{}^{m}/_{s^2}\,(30m)}{0.0252\frac{400m}{0.1m} + 0.5 + 118 + 1}} = 1.635\,{}^{m}/_{s}$$

5. Calculate Reynolds number

$$Re = \frac{\rho V_{avg} D_h}{\mu}$$

$$Re = \frac{\left(998\,{}^{kg}/_{m^3}\right)(1.635\,{}^{m}/_{s})(0.1m)}{1.01e^{-3} Pa \cdot s} = 161{,}514$$

6. Calculate friction factor (f):

Use Haaland Equation with $\varepsilon = 0.0026$ and $Re = 161{,}514$

$$f = \left[\frac{1}{-1.8\log_{10}\left[\left(\frac{0.0026}{3.7}\right)^{1.11} + \frac{6.9}{161{,}514}\right]}\right]^2 = 0.0260$$

Since our new value for f is different than our guessed value, we need to iterate!

7. Calculate new average velocity using Bernoulli Equation:

$$V_{avg} = \sqrt{\frac{2 * 9.81\,{}^{m}/_{s^2}\,(30m)}{0.0260\frac{400m}{0.1m} + 0.5 + 118 + 1}} = 1.623\,{}^{m}/_{s}$$

8. Calculate new Reynolds number

$$Re = \frac{\left(998\,{}^{kg}/_{m^3}\right)(1.623\,{}^{m}/_{s})(0.1m)}{1.01e^{-3}Pa \cdot s} = 160{,}346$$

9. Calculate new friction factor (f):

Use Haaland Equation with $\varepsilon = 0.0026$ and $Re = 160{,}346$

$$f = 0.0260$$

Since our new value for f is same as our previous value, the average velocity is $1.623\,{}^{m}/_{s}$!

10. Calculate volumetric flow rate (Q):

$$Q = V_{avg}A = V_{avg}\frac{\pi}{4}D^2 = 1.623\,{}^{m}/_{s}\frac{\pi}{4}(0.1m)^2 = 0.0127\frac{m^3}{s} * \frac{60s}{1min}$$

$$\boldsymbol{Q = 0.765\frac{m^3}{min}}$$

b) Fill in the table below by redoing the calculation from part a) for the four other butterfly valve disc positions.

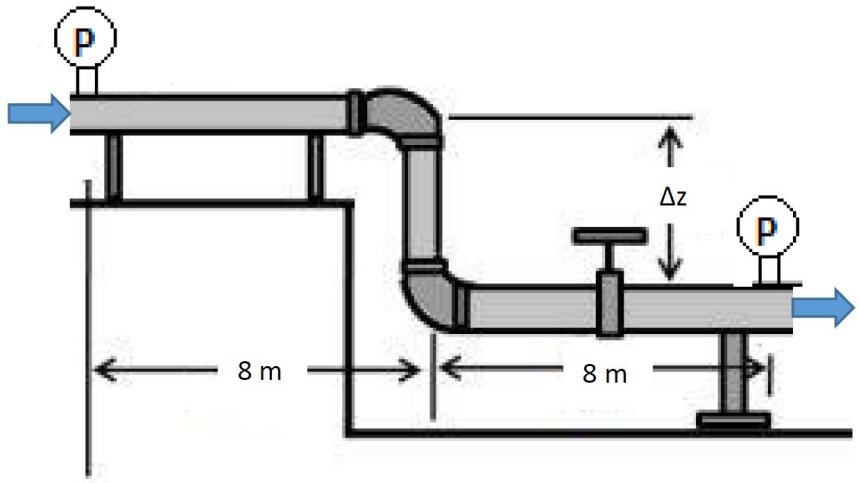
Valve Position	Representative Loss Coefficient From Slide 19 of PDF_08	Volumetric flow rate of water (m^3/min)
θ = 60°	K = 118	0.765
θ = 40°	K = 10.8	1.064
θ = 20°	K = 1.54	1.110
θ = 10°	K = 0.52	1.115
θ = 5°	K = 0.24	1.117

c) Compare your answers from part b) to that which we determined in lecture for the same system without a butterfly valve.

QUESTION 3

Glycerin at 20 °C, a very viscous fluid (1.4 Pa·s), is being transported to a membrane production line through the piping system shown in the figure below (not drawn to scale) with the following details:

- All the piping is 4" schedule 40 made of stainless steel
- Both elbows are 90o threaded regular (K = 1.4)
- The hand operated globe valve (on lower portion of piping) is fully open (K = 10)



a) Write the simplified form of the Bernoulli equation that describes the transport of glycerin from the pressure gauge on the upper portion of piping down to the pressure gauge on the lower portion of piping.

$$\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + gz_1 = \frac{P_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + gz_2 + h_l + h_{l_m}$$

Since the pipe has same diameter, by conservation of mass:

$$V_1 = V_2$$

Thus, the simplified form of the Bernoulli Equation is:

$$\frac{P_1}{\rho} + gz_1 = \frac{P_2}{\rho} + gz_2 + h_l + h_{l_m}$$

or

$$\frac{P_1}{\rho} + gz_1 = \frac{P_2}{\rho} + gz_2 + f\frac{L}{D_h}\frac{V_{avg}^2}{2} + \sum\left[K\frac{V_c^2}{2}\right]$$

$$\frac{P_1}{\rho} + gz_1 = \frac{P_2}{\rho} + gz_2 + f\frac{L}{D_h}\frac{V_{avg}^2}{2} + 2K_{elbow_{threaded-regular}}\frac{V_{avg}^2}{2} + K_{globe-valve}\frac{V_{avg}^2}{2}$$

b) Determine the elevation change (m) between the upper and lower portions of the piping system (which is indicated on the figure as Δz) given that the differential pressure between the first pressure gauge and the second pressure gauge (i.e. P1 – P2) is 1500 kPa when the volumetric flow rate of glycerin is 80 L/s.

As shown on the figure above, the total length of piping (L) depends on the elevation change:

$$L = 16 + \Delta z$$

Thus, the Bernoulli Equation is now:

$$\frac{P_1}{\rho} + g\Delta z = \frac{P_2}{\rho} + f\frac{(16+\Delta z)}{D_h}\frac{V_{avg}^2}{2} + 2K_{elbow_{threaded-regular}}\frac{V_{avg}^2}{2} + K_{globe-valve}\frac{V_{avg}^2}{2}$$

which can be rearranged to the following:

$$\Delta z = \frac{\frac{-(P_1 - P_2)}{\rho g} + \frac{V_{avg}^2}{2g}\left[f\frac{(16)}{D_h} + 2K_{elbow_{threaded-regular}} + K_{globe-valve}\right]}{\left(1 - f\frac{1}{D_h}\frac{V_{avg}^2}{2g}\right)}$$

Known values in above equation:

$P_1 - P_2 = 1500000\ Pa$

$\rho = 1260\ {}^{kg}/_{m^3}$ ←From Table A2 (on Slide 22 of PDF_00)

$D_h = 0.1023m$ ←From Table on Slide 51 of PDF_07

$$V_{avg} = \frac{Q}{A} = \frac{Q}{\frac{\pi}{4}{D_h}^2} = \frac{80\ {}^{L}/_{s}\frac{1m^3}{1000L}}{\frac{\pi}{4}(0.1023m)^2} = 9.73\ {}^{m}/_{s}$$

Need to determine the friction factor (f), which requires knowing the Reynolds number:

$$Re = \frac{\rho V_{avg} D_h}{\mu}$$

$$Re = \frac{\left(1260\ {}^{kg}/_{m^3}\right)(9.73\ {}^{m}/_{s})(0.1023m)}{1.4Pa \cdot s} = 896$$

Since Re < 2300, the flow is laminar and so:

$$f = \frac{64}{Re} = 0.071$$

Substituting in values given above:

$$\Delta z = \frac{\frac{-(1500000Pa)}{\left(1260\ {}^{kg}/_{m^3}\right)\left(9.81\ {}^{m}/_{s^2}\right)} + \frac{(9.73\ {}^{m}/_{s})^2}{2\left(9.81\ {}^{m}/_{s^2}\right)}\left[0.071\frac{(16m)}{0.1023m} + 2(1.4) + 10\right]}{\left(1 - 0.071\frac{1}{0.1023m}\frac{(9.73\ {}^{m}/_{s})^2}{2\left(9.81\ {}^{m}/_{s^2}\right)}\right)}$$

$$\boldsymbol{\Delta z = 2.37m}$$